

CLASS 11TH BIOLOGY - SECTION A (BOTANY)

ANNUAL BOARD PAPER

Series: 11thARNKD(W/Z) JKLUT-25-1210-A Time: 1.5 Hours

Max. Marks: 35 Marks

Note: Attempt any 30 Marks out of 35 Marks. | Illustrate with labeled sketches where required.

SECTION A: OBJECTIVE TYPE QUESTIONS

1 Mark each

- Q1. As we move from species to kingdom in a taxonomic hierarchy, the number of common characteristics:
- (A) Will decrease (B) Will increase
(C) Remain same (D) May increase or decrease
- Q2. Eukaryotic unicellular organisms belong to Kingdom:
- (A) Monera (B) Protista (C) Fungi (D) Bacteria
- Q3. The arrangement of flowers on a floral axis is called:
- (A) Aestivation (B) Inflorescence (C) Phyllotaxy (D) Placentation
- Q4. Casparian strips are impermeable suberized bands found in:
- (A) Epidermis (B) Hypodermis (C) Endodermis (D) Pericycle
- Q5. Chloroplast dimorphism (Kranz anatomy) is a characteristic feature of:
- (A) All plants (B) Only in Algae (C) C3 plants (D) C4 plants

SECTION B: VERY SHORT ANSWER TYPE QUESTIONS

2 Marks each

- Q6. What do you understand by Taxonomical Hierarchy? List the categories in ascending order.
- Q7. Define Natural classification system. Who proposed it?
- Q8. Differentiate between Green Algae (Chlorophyceae) and Brown Algae (Phaeophyceae) on the basis of photosynthetic pigments and stored food materials.
- Q9. Give any two structural differences between the vascular bundles of monocot stem and dicot stem.
- Q10. Differentiate between Endarch and Exarch primary xylem development conditions.

SECTION C: SHORT ANSWER TYPE QUESTIONS

3 Marks each

- Q11. Write down the universal international rules for Binomial Nomenclature as governed by ICBN.
- Q12. What are the salient morphological and cellular features of Kingdom Monera?
- Q13. What primary criteria were used by R.H. Whittaker to classify living organisms into the Five Kingdom system?
- Q14. Define Respiratory Quotient (RQ). What is its numerical value for carbohydrates, fats, and organic acids?
- Q15. Name various plant growth promoters (phytohormones). List the major physiological uses of any one of them.

SECTION D: LONG ANSWER TYPE QUESTIONS

5 Marks

Q16. Give a comprehensive schematic account of Glycolysis (EMP Pathway). Where does it occur in the cell?

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Describe the process and Z-scheme of non-cyclic photophosphorylation in higher plants with an energy diagram.

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